Animesh Banik

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Research Interests

- Quantum Information
- Quantum Simulation
- Quantum Communication
- Quantum Error Correction
- Quantum Algorithms
- Quantum Computing

Publications

1. **Animesh Banik**, Md. Shihab Khan, Rafid Masrur Khan, Syed Emad Uddin Shubha, Mahdy Rahman Chowdhury, "Secure and Efficient n-Qubit Entangled State Teleportation Using Partially Entangled GHZ Channels and Optimal POVM". AVS Quantum Science, 2025 (Q1, IF: 3.0).

   `doi: 10.1116/5.0284072  Github link`

2. **Animesh Banik**, Md. Shihab Khan, Rafid Masrur Khan, Syed Emad Uddin Shubha, Quazi Muhammad Rashed Nizam, "Generalized State Discrimination for Tunable Quantum Key Distribution: The phiQKD Protocol". ArXiv Preprint

   `doi: 10.48550/arXiv.2511.06488  Github link`

Conference Presentations

1. **Animesh Banik**, Md. Shihab Khan, Rafid Masrur Khan, Syed Emad Uddin Shubha, Quazi Muhammad Rashed Nizam, "Efficient Quantum Key Distribution using Generalized State Discrimination" 6th International Conference on Physics (ICP-2026), organized by Bangladesh Physical Society, University of Dhaka, Bangladesh.

2. **Animesh Banik**, Md. Shihab Khan, Rafid Masrur Khan, Syed Emad Uddin Shubha, Quazi Muhammad Rashed Nizam, "Generalized State Discrimination for Binary Non-orthogonal States." 6th International Conference on Physics for Sustainable Development and Technology (ICPSDT-2025), Chittagong University of Engineering & Technology (CUET), Bangladesh.

3. **Animesh Banik**, Md. Shihab Khan, Md. Tareq Aziz, Dr. Quazi Muhammad Rashed Nizam. "In search of a potential method for teaching quantum mechanics at undergraduate level". In 5th ICPSDT organized by Department of Physics, CUET.

Research & Teaching Experience

Research Assistant - QRNLab — July 2023 - Present

Supervisor: Professor Dr. Quazi Muhammad Rashed Nizam

- Designed and delivered lecture materials for graduate-level Quantum Mechanics courses, improving clarity and engagement for 90+ students.
- Assisted in course restructuring by integrating problem sets and simulations, enhancing student problem-solving practice.
- Co-supervised undergraduate researchers, guiding them toward independent project development and manuscript preparation.
- Conducting research in quantum communication and simulation, leading multiple manuscripts under review and preparation.

- **Ongoing Research Projects:**
  1. "Quantum Simulation of UV-Vis Spectroscopy – Utilizing VQE and SSVQE to calculate transition dipole moments and continuous absorption spectra for complex molecular systems" (Manuscript in preparation).
  2. "Simulating Holographic Scrambling and the Information Paradox" - Formulating optimal POVM-based decoding protocols to model entanglement dynamics and information recovery in black hole toy models.

Education

University of Chittagong — January 2025 – April 2026
M.S. in Physics — GPA: 3.48/4.00

University of Chittagong — January 2019 - December 2024
B.Sc. in Physics — CGPA: 3.40/4.00

Professional Skills

Programming Languages	: Python, C, C++.
Simulation Softwares	: Qiskit, PennyLane, Cirq.
Formatting Tool	: LaTeX.

Honors and Awards

- *National Science and Technology Fellowship (2025-2026)* - Awarded for academic merit and research potential.
- *Global Quantum Mechanics Challenge 2026* - Qualified for the Semi-finals, reflecting scientific problem-solving skills and interest in quantum mechanics beyond the institutional curriculum.
- *Best Oral Presenter* - 6th International Conference on Physics for Sustainable Development and Technology (ICPSDT-2025), CUET, Bangladesh. Recognized by the University of Chittagong International Affairs Office and CURHS (2026) for this achievement.
- QWorld Diplomas (*QSilver*, *QBronze*) - Awarded for completing the QSilver-35 and QBronze-167 Workshops organized by QWorld and respective QCousins.
- International *Physics*, *Astronomy and Astrophysics* Olympiads - Qualified for the Semi-final Round (Physics 2025) and Pre-final Round (Astronomy 2023), showcasing strong skills in physics, astronomy, and analytical reasoning.

Selected Projects & Hackathons

- *MIT iQuHACK 2025 (IonQ Challenge)* - Developed and submitted a quantum computing project utilizing IonQ hardware.
- *2nd Place, Track C - Q-volution Hackathon (2026)* - Awarded to my team "PlanQtons" in the global quantum hackathon organized by Girls in Quantum. `Github link`
- *Quantum State Discrimination* - Developed Qiskit implementations of minimum-error (Helstrom) and unambiguous state discrimination protocols using POVMs and Naimark dilation. `Github link`
- *Quantum Random Number Generator* - Implemented a quantum random number generator in Qiskit using single-qubit superposition and measurement. `Github link`

Professional Development

- *IBM Quantum Excellence Badge (QGSS 2025)* - Awarded for completing all four core labs of the IBM Quantum Global Summer School 2025.

- *QCourse 581-1: Selected Topics in Quantum Science and Technologies (2026)* - Successfully completed a 3 ECTS university-accredited course organized by QWorld in collaboration with the Department of Computing, University of Latvia. Studied Quantum Error Correction (QEC), Quantum Key Distribution (QKD), and Quantum Machine Learning (QML) through lectures, coursework, and assessments.
- *Aspire Leaders Program (Cohort 3, 2026)* - Selected for a fully funded international leadership development program for high-potential university students.
- *Wiser Summer Program 2026: What is the Future of Optimization? (Classical + AI + Quantum)* - Successfully completed an international program on classical, AI, and quantum optimization.
- *QWorld QTraining for Bronze (Instructor)* - Successfully completed QTraining10-3 and served as an instructor in a mock QBronze workshop organized by QWorld (September 2025).
- *Mahdy Research Academy Thesis Program (2024–2025)* - Completed both (*first* and *second*) parts of the private online thesis course/program in Quantum Computing & Information.

Leadership & Outreach

- **Team Leader, QRNLab Outreach Booth:** Led the QRNLab team in organizing and managing a public outreach booth at a national research festival, engaging hundreds of visitors with demonstrations on quantum physics and fostering scientific curiosity among students and the general public.
- **Qiskit Advocate:** Engaged in global community outreach, mentoring beginners, and contributing educational content to promote quantum computing literacy through IBM Quantum initiatives.
- **QBangladesh Member:** Author monthly science communication articles for the QBangladesh newsletter, promoting quantum literacy to the national physics community through “QTalk” seminars.
- **Mentor, QBronze190 (QWorld):** Provided technical guidance and hands-on support for international students, focusing on Qiskit implementation and debugging of foundational quantum algorithms.

Test Scores

- **IELTS Academic (2025)**: Overall 8.0/9.0 [Reading: 9.0; Listening: 8.5; Writing: 8.0; Speaking: 7.0]